

CNN Model Activity

Wall (2019) discusses the uses and risks of facial recognition technology.

The author highlights several critical aspects of using this tool for security matters, such as failing to correctly recognize the right person and biases arising from the data used to train the model.

From an ethical perspective, the use of CNN-based systems for public security presents several issues. One of the principal problems is the reliability of the system. As highlighted in the article, AI models can make several mistakes, such as confusing a muffin with a Chihuahua. The use of CNN-based systems for surveillance purposes can have serious consequences, such as incorrect identification, unfair accusations, or wrongful arrests.

Another issue is the bias present in the training data. If the dataset is not balanced or representative, the systems can show worse performance on specific social or ethnic groups, contributing to forms of algorithmic discrimination.

In this context, it is important to consider the topic of responsibility: when these systems make a mistake, it is not clear who should be held responsible for the damage caused. In addition, questions regarding data collection and storage remain: there is no guarantee that the collected data will be used exclusively for security purposes or that it will not be shared with third parties.

Moving on to the privacy aspect, the fact that the image of a person can be captured and stored while they are in a public space can highlight important concerns, mostly in

the case of data breaches. A data breach could lead to the unauthorized disclosure of personal information.

Finally, the widespread adoption of AI-based surveillance systems can have a significant impact on society. The awareness of being constantly monitored could change people's behavior in public spaces, reducing the freedom of social interaction. In addition, continuous government surveillance raises concerns about the balance between security and individual freedom.

Reference list

Wall, M (2019) Biased and wrong? Facial recognition tech in the dock. BBC News, 6 July. Available at: <https://www.bbc.com/news/business-48842750>

CNN Model

In this exercise, the goal was to change the index 16 with a number between 1 and 15 and observe whether the model predicts the correct class. I decided to make two changes:

- I changed index 16 to index 4. The true label of the selected image was "frog" while the CNN predicted the class "deer".
- I changed index 16 to index 7. The selected image was labelled as "frog," and the CNN correctly predicted the same label.

The two changes suggest that the CNN can correctly classify some images after training but may misclassify others depending on their visual characteristics.